

Obaid Ullah Khan

+92-3446286836 | Email | LinkedIn | GitHub | Portfolio

SUMMARY

AI and Machine Learning Engineer with hands-on experience in training, optimizing, and deploying classical ML and deep learning models, as well as building LLM-powered Retrieval-Augmented Generation (RAG) systems. Proven ability to design end-to-end production pipelines, from data preprocessing and model inference to scalable API and vector search services, across legal, financial, and energy analytics domains using Python, Django, FastAPI, Terraform, CI/CD, and modern MLOps practices.

EDUCATION

University of Malakand

Bachelor of Science in Computer Science

Pakistan

2021 - 2025

EXPERIENCE

AI Engineer

Sep 2024 - Present

Data Function Inc., Islamabad

- Architected a RAG-based Legal AI Assistant (CasePolaris) using LangChain, Vector Databases, and LLMs for semantic retrieval and legal reasoning over Canadian court cases.
- Designed and maintained scalable Django REST APIs for enterprise ERP (StoneCloud), supporting finance, logistics, and reporting workflows.
- Built real-time solar monitoring and analytics pipelines (ForceV) with automated energy computations and visualization dashboards.
- Developed marketing attribution models to measure YouTube Shorts vs. traditional ads, enabling data-driven ROI optimization.
- Created reusable Terraform modules for provisioning and managing Azure resources across projects.
- Built and maintain a central GitHub Actions repository providing org-wide CI/CD pipeline templates, called by consumer projects instead of each writing its own CI/CD pipeline from scratch.

Machine Learning Intern

Jun 2024 - Sep 2024

IT-Solera, Islamabad

- Trained and fine-tuned deep learning models for NLP and Computer Vision using TensorFlow and PyTorch.
- Applied data augmentation and feature engineering to improve model performance on limited datasets.

PROJECTS

World Wars RAG Chatbot | *LangChain, OpenAI, FAISS, Streamlit*

- Built a historical question-answering system using Retrieval-Augmented Generation over large document collections.
- Implemented semantic search with FAISS and contextual response generation using LLMs.

Credit Card Fraud Detection | *Python, XGBoost, Scikit-Learn*

- Trained ensemble models on highly imbalanced financial datasets using SMOTE and advanced feature engineering.
- Optimized precision-recall tradeoff to minimize false positives in real-world fraud scenarios.

TECHNICAL SKILLS

AI/ML: Python, TensorFlow, PyTorch, Scikit-learn, NLP, RAG, LLMs (OpenAI), LangChain, Pinecone

Backend: Django, FastAPI, Flask, REST APIs, MySQL

Cloud & DevOps: Azure, Terraform, CI/CD, Docker, Git, AWS (Basic)

Tools: Streamlit

Languages: Python, SQL